Portuguese Legislative and Regional Elections

Voting Patterns and Political Shifts

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1 Introduction

Project Overview

Objective: End-to-end design and implementation of a system for managing and analysing Portuguese electoral data.

Core Components:

Normalised relational database (PostgreSQL).

Data warehouse for analytical queries.

Custom ETL pipeline for official government data.

Spatial database capabilities (PostGIS).

Lightweight functional frontend (OCaml).

2 Data Collection and ETL

Scope and Standardization

Dataset: Local elections (2017, 2021, 2025) and Legislative elections (2015, 2019, 2022, 2024, 2025).

Standardization: Older .xls converted to .xlsx.

ETL Strategy:

Separated local vs. legislative parsing.

Parsed by territory, office, and matrix layouts.

Bypassed staging tables in favor of strong parser validation and database triggers.

Normalization and Debugging

Triggers: Ensured derived summaries stayed consistent after data ingestion.

Normalization: Territory codes canonicalized (e.g., 6-digit parish codes mapped appropriately). Party/Coalition labels standardized.

Spatial Data: Geometries linked to territories (e.g., Azores/Madeira circles derived from island districts).

3 Database Architecture

Two-Layer Design

Operational Schema (op)

The central source of truth. Highly normalized. Handles raw imports, territorial hierarchies, and official vs. calculated seat methods (D'Hondt).

Warehouse Schema (wh)

Refreshed from op. Uses dimension and fact tables (Turnout, Votes, Seats) specifically designed for fast analytical querying and visualization.

4 Data Treatment and Analysis

Analytical SQL Functions

Territory Resolution: wh.resolve_vote_territory dynamically matches user selection to the correct voting level (Circle, Municipality, Parish).

Party Aggregation: wh.party_results handles roll-ups, seat differences, and canonical entity grouping.

Abstention and Swing: Dedicated functions to calculate geographic abstention rates and Left/Right vote swings across elections.

5 Application Layer

Frontend and Map Generation

Backend: Lightweight API routing requests to the PostgreSQL warehouse.

Frontend: Developed in OCaml, featuring interactive maps and dynamic charts.

Plotting Micro-service: A Python API that requests to the PostgreSQL warehouse to build plots, passes them to the backend to be later displayed in the frontend.

Frontend and Map Generation (ii)

6 Results

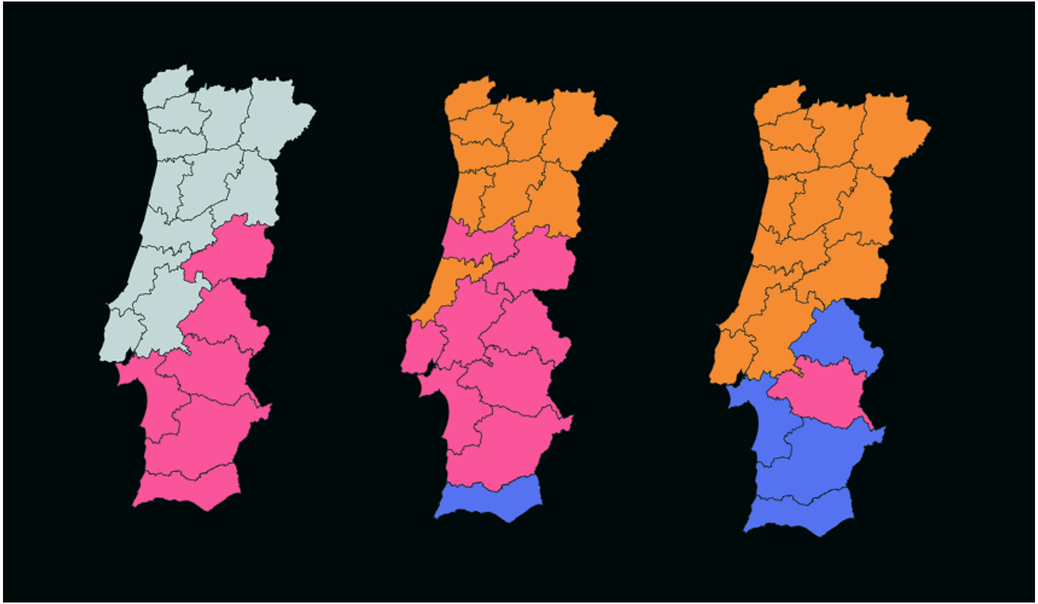
Electoral Realignment in the South

Data visualizations revealed a clear electoral realignment in southern Portugal.

Historical dominance of left-wing parties (e.g., Beja, Évora) showed marked erosion in 2024 and 2025.

Demonstrates a notable vote swing toward right-wing and alternative political forces.

Electoral Realignment in the South (ii)



Conclusion

Moving complex electoral counting rules and spatial rendering into the database (wh schema) resulted in a thin, performant API.

The system reliably normalizes raw government spreadsheets into rich, drillable analytical insights.